MILITARY SPECIFICATION

BOX, PATHOLOGICAL SPECIMEN, SHIPPING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a reuseable controlled temperature shipping container sufficiently durable for use in air and parcel post shipment of anatomical specimen or other items requiring closely controlled temperatures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390C	- Plastic, Molding And Extrusion Material, Polyethylene and Copolymers (Low Medium and High Density).
PPP-B-636B	- Boxes, Fiberboard.

MILITARY

MIL-F-21840E	- Fastener, Tapes, Hook and Pile, Synthetic.
MIL-P-116	- Preservation Packaging, Methods Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this codument should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, CODE ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8115

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. The material shall be suitable in every respect for the purpose intended and shall be as specified herein.

3.1.1 High density polyethylene. High density polyethylene material shall conform to L-P-390C, type 1, grade 4, class H, category 3.

3.1.2 Freon blown polyurethane. Freon blown polyurethane core specimen shall have the following properities:

Core density of 2.05 ± 0.1 lb per cu ft (free rise).
Closed cell content of 90% min.
Compression strength at yield of 30 psi.
Thermal conductivity (K-Factor) of .12 BTU-in/hr-ft^2-^{o}F at 75°F.

3.1.3 Cross linked expanded polyethylene. The cross linked expanded polyethylene shall have a closed cell structure with the following properties:

Bulk density of 2 lbs per cu ft.
Compression strength of 30 to 40 psi in the cross direction at 20°C.
Permanent set of 4.9 percent (at 25 percent compression).
Thermal conductivity (K-Factor) of 0.278 BTU-in/hr-ft^2-^{o}F at 75°F.

3.2 Design and construction. The pathological specimen shipping box shall consist of a container, gasket, hinged cover, carrying strap and sealing fitting. See Figure 1 for a typical design of the shipping box. Interior box shall be smooth with no protrusions or joints except at seal.

3.2.1 Container. The container shall be fabricated of a high density polyethylene shell filled with freon blown polyurethane. The interior dimensions shall be 17 ± 1/4 inches long, 6-1/4 ± 1/4 inches wide and 7-1/2 ± 1/4 inches deep. A plastic latching mechanism shall be provided

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

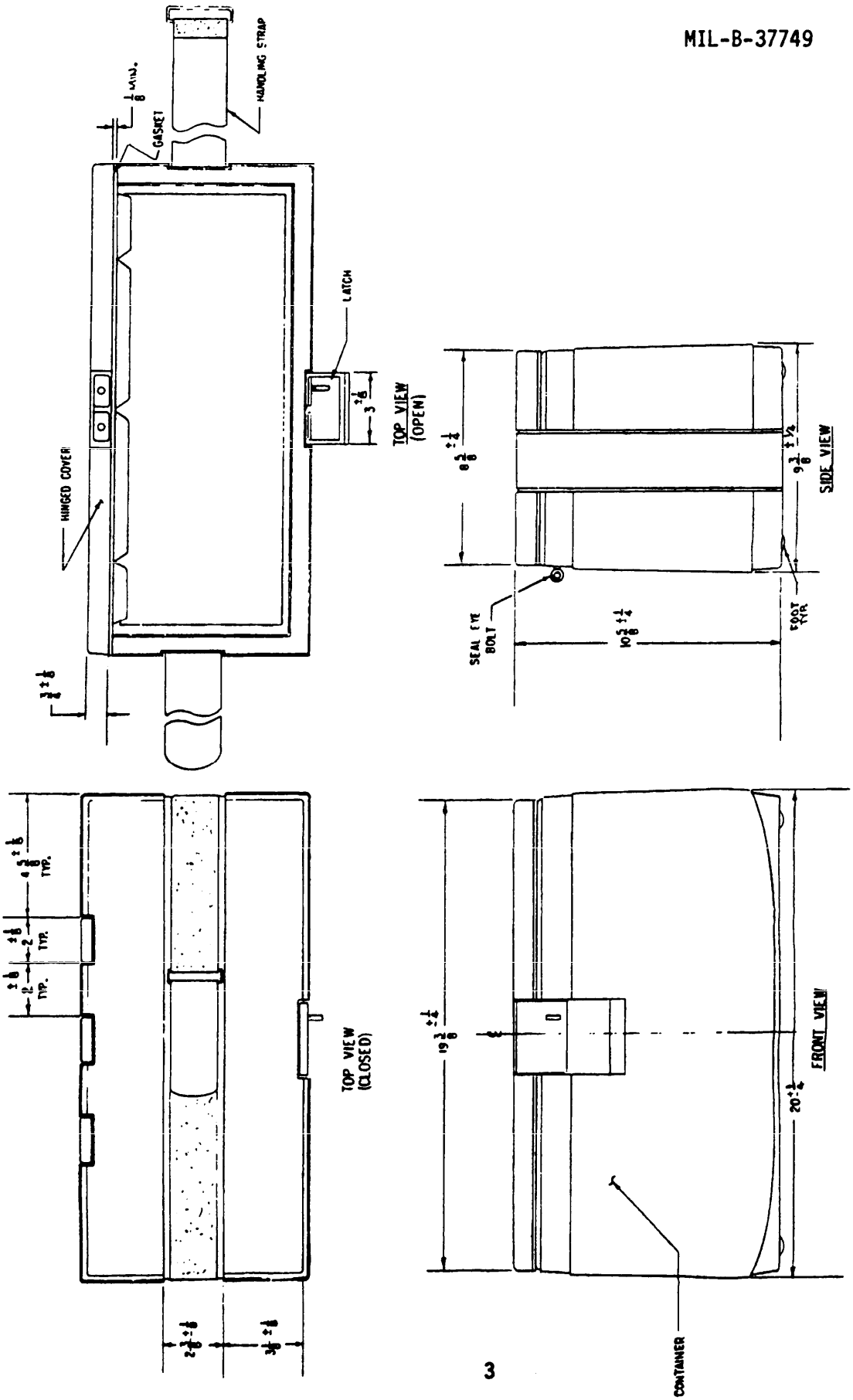


FIGURE 1. Box, Pathological Specimen, Shipping

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

on the front of the container to secure the hinged cover (3.2.3) in the closed position. The outside of the container shall contain a 2-3/8 inch ± 1/8 inch wide by 1/8 inch (min.) deep vertical groove in the center of each end surface. The carrying strap (see 3.2.4) shall recess in this groove. Four feet, one in each corner, shall be provided on the bottom of the container. Each foot shall be 1/8 inch (min.) high.

3.2.2 Gasket. The gasket shall be fabricated of cross linked expanded polyethylene foam not less than 1/8 inch thick. The gasket shall fit the container and hinged cover interface and shall be attached to the cover with suitable adhesive.

3.2.3 Hinged cover. The hinged cover shall consist of a high density polyethylene shell filled with freon blown polyurethane foam. The cover thickness shall be 1-1/2 ± 1/4 inches. The cover shall be attached to the container by means of corrosion resistant hinges or hinge pins and shall provide a corrosion resistant latching fitting to allow the cover to be secured to the container. A 2-3/8 ± 1/8 inch wide by 1/8 inch (min.) deep continuous groove shall be provided in the longitudinal center of the top and ends to recess the carrying strap. The closure system shall ensure that a constant pressure is exerted between the cover and the container.

3.2.4 Carrying strap. The carrying strap shall consist of a woven nylon strap and a "D" ring. The finished strap shall be 2 ± 1/8 inches wide by 65 ± 1 inches long including the "D" ring which shall be suitably attached to one end of the strap. The entire outer surface of the strap shall contain hook type fasteners manufactured in accordance with type I, class 1 of MIL-F-21840E except for the last 2-1/2 ± 1/8 inches of the end opposite the "D" ring. This portion of the strap shall contain loop type fasteners manufactured in accordance with class 1 of MIL-F-21840E. The strap shall recess in the grooves provided in the container and closed cover when the strap is secured. The strap shall be suitably attached to the bottom of the container to position the "D" ring at the center of the closed cover with the strap secured. The "D" ring shall be fabricated of corrosion resistant metal. The carrying strap shall enable the box to be carried by one hand.

3.2.5 Sealing fitting. The sealing fitting shall consist of a 1/8 inch (min.) inside diameter eyebolt fabricated of corrosion resistant metal. The eyebolt threaded section shall be securely fastened to the container to enable the eye to protrude through a slot in the latch moveable portion when the latch is secured to the cover.

3.3 Weight. The pathological specimen shipping box weight shall be 10 ± 1/2 lbs.

3.4 Cleaning. The interior and exterior surfaces of the pathological specimen shipping box and carrying strap shall be completely washable and capable of being cleaned repeatedly with warm soapy water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

3.5 Water leakage. The pathological specimen shipping box shall show no visible signs of water leakage when tested as specified (see 4.4.1).

3.6 Air leakage. The pathological specimen shipping box shall show no visible signs of leakage when tested as specified (see 4.4.2).

3.7 Temperature tests.

3.7.1 Refrigeration test. When tested as specified in 4.4.3 the payload shall be maintained at a temperature from 2 to 8°C after temperature stabilization has been reached. Total time for temperature stabilization and maintaining temperature from 2 to 8°C shall be not less than 72 hours.

3.7.2 Freezing test. When tested as specified in 4.4.4 the payload shall be maintained at a temperature from -10 to -20°C after temperature stabilization has been reached. Total time for temperature stabilization and maintaining from -10 to -20°C temperature shall be not less than 72 hours.

3.8 Transportation tests.

3.8.1 Vibration test. The pathological specimen shipping box shall show no visible signs of damage and the latch or strap shall not open when tested as specified (see 4.4.5.1).

3.8.2 Shock test. The pathological specimen shipping box shall show no visible signs of damage and the latch or strap shall not open when tested as specified (see 4.4.5.2).

3.9 Finish. The exterior surface of the pathological specimen shipping container shall be capable of having labels affixed that comply with Title 42 Public Health Act as revised 1 October 1973.

3.10 Identification marking. The pathological specimen shipping container shall be permanently and legibly marked with the manufacturer's name or registered trademark.

3.11 Instructions for use. The pathological specimen shipping container shall be provided with instructions for use which shall include a caution statement indicating that prior to shipping, a label complying with Title 42 Public Health Act (Etiologic Agents) Biomedical Material shall be attached to the container, exterior top or side surfaces.

3.12 Workmanship. Workmanship shall be first class throughout and all components shall be free from defects which detract from their appearance or impair their serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

3.13 Medical devices requirements. The pathological specimen shipping container shall be in accordance with applicable requirements of the Bureau of Medical Devices and Diagnostics.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. Unit of product shall be one pathological specimen shipping box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Visual examination		
Major defects	II	1.0
Minor defects	II	2.5
Dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Unit of product for testing shall be one pathological specimen shipping box.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Water leakage test	3.5	4.4.1	S-1	*
Air leakage test	3.6	4.4.2	S-1	*
Refrigeration test	3.7.1	4.4.3	S-1	*
Freezing test	3.7.2	4.4.4	S-1	*
Vibration test	3.8.1	4.4.5.1	S-1	*
Shock test	3.8.2	4.4.5.2	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The pathological specimen shipping box shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III.

TABLE III. Classification of defects.

Categories	Defects*
Major	
101	Box not free of cuts, tears or deformation.
102	Cover does not mate with container.
103	Parts fastened with adhesive do not completely adhere.
104	Cover of box not stationary in an upright position.
105	Instructions not present, correct or legible.
106	Gasket or carrying strap not free of cuts, tears, or holes.
Minor	
201	Finish not free of scratches or scuff marks.
202	Finish not smooth, free of runs, drops or streaks.
203	Finish, if colored, not free of discolored areas or areas not colored.
204	Manufacturer's identification marking not present, complete, correct, legible or permanent.

* Examination shall not be restricted to the classified defects listed above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

4.3.2 Dimensional examination. The shipping box shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Water leakage test. The pathological specimen shipping box shall be filled with water, the cover closed and latched and the strap secured. The shipping box shall be turned upside down and set on its top on a flat surface. The box shall be left undisturbed for a period of 5 minutes prior to examination for leakage.

4.4.2 Air leakage test. The latched pathological specimen shipping box with the strap secured shall be pressurized internally with 2 psi gage air pressure and completely submerged in a water bath for a period of not less than 2 minutes.

4.4.3 Refrigeration test. A plastic container filled with 10 pounds of water (payload) shall be placed in the pathological specimen shipping box with suitable activated temperature controllant canisters (see 6.3) rated to maintain a payload temperature from 2 to 8°C. Temperature monitoring equipment shall be attached to the payload, the cover closed and latched, the strap secured, and the payload temperature monitored for 72 hours. The external environment (room temperature) of the box shall be maintained at 2 ± 3°C during the test.

4.4.4 Freezing test. A plastic container filled with 10 pounds of frozen water (payload) shall be placed in the pathological specimen shipping box with suitable activated temperature controllant canister (see 6.3) to maintain a payload temperature form -10 to -20°C. Temperature monitoring equipment shall be attached to the payload, the box cover closed and latched, the strap secured, and the payload temperature monitored for 72 hours. The external environment (room temperature) of the box shall be maintained at 22 ± 3°C during the test.

4.4.5 Transportation tests.

4.4.5.1 Vibration test. The pathological specimen shipping box loaded as specified in 4.4.3, with the cover closed and latched and the strap secured, shall be subjected to the vibration tests specified in MIL-STD-810, Method 514.2, Procedure XI, Equipment transported as loose cargo.

4.4.5.2 Shock test. The pathological specimen shipping box loaded as specified in 4.4.3, with the cover closed and latched and the strap secured, shall be subjected to shock tests specified in MIL-STD-810, Method 516.2, Procedure

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

II, Transit drop test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each box shall be thoroughly cleaned in accordance with MIL-P-116, method C-1.

5.1.1.2 Drying. Each box shall be thoroughly dry.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Each box shall be packaged in accordance with MIL-P-116, method III, in a shipping container as specified in 5.2.

5.1.1.5 Level C. Each box shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Each pathological box shall be packed in a shipping container designed for a type I load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Each pathological box shall be packed in a shipping container designed for type I load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. Pathological boxes packaged as specified in 5.1.1.5 shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking. Marking shall be as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The item covered by this specification is a controlled temperature shipping container whose intended use is to ship anatomical specimens or other items requiring closely controlled temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Temperature controllant canisters may be obtained from KWIK-KOLD DIVISION KAY LABORATORIES, INC., P.O. Box 696, Moberly, MO 65270.

6.4 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet military requirements.

6.5 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
8115-01-013-8533	BOX, PATHOLOGICAL SPECIMEN, SHIPPING, Insulated

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 8115-0416

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:25Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-B-37749 BOX, PATHOLOGICAL SPECIMEN, SHIPPING

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

POSTAGE AND FEES PAID

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

DEFENSE PERSONNEL SUPPORT CENTER
DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:25Z
Check the source to verify that this is the current version before use.